

Introduction to Programming (CSE141)

Syllabus

Fall'24

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1 Basic Information

Credits	3 theory + 1 lab hours
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1.1 Objectives

This course serves as an introduction to computational problem solving using a modern programming language. The main objective is to provide students with understanding of programming essentials, including algorithms, data types, elementary control structures, functions, and recursion used within the framework of structural and modular programming paradigms. The students will learn to apply an integrated program development tool to design, implement, test, debug, and document programs. This course prepares students for the more advanced programming courses such as CSE142 (Object Oriented Programming Techniques) and CSE247 (Data Structures).

2 Course Contents

We will cover following topics:

- Introduction to programming in C++: Elementary data types, control flow, conditionals, loops, and arrays.
- Standard Input and output at console, file I/O, as sounds, and as on-screen drawings.
- Pointers, references, dynamic memory allocation.
- Functions, pass by value/reference.
- Libraries, modular programming.
- Recursion.
- User-defined datatypes, classes, objects, and methods.

2.1 Weekly Schedule (tentative)

Table 1: Weekly schedule of topics and assessments

	Topic(s)	Notes
Week 1 <i>Aug 19 – Aug 25</i>	Introduction C++ as first programming language First program: “Hello World” Terminal, IDE, Input from command-line	

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Table 1: Weekly schedule of topics and assessments (Continued)

	Topic(s)	Notes
Week 2 <i>Aug 26 – Sep 1</i>	Built-in Data Types Variables, Declaration & Assignment statements Integer data type: <code>int</code> , <code>long</code> , <code>short</code> Floating-point numbers: <code>double</code> , <code>float</code> <code>std::string</code> data type <code>bool</code> data type, Comparison operators	Homework 1 out
Week 3 <i>Sep 2 – Sep 8</i>	Conditionals & Loops Type conversion <code>if</code> , <code>while</code> , and <code>for</code> statements Example: <i>Newton-Raphson</i> method Nested <code>if-else-if</code> conditions	Homework 1 due
Week 4 <i>Sep 9 – Sep 15</i>	Nested Loops Examples: Gambler's Ruin, Factoring integers <code>do-while</code> loop, <code>break</code> and <code>continue</code> statements	Homework 2 out
Week 5 <i>Sep 16 – Sep 22</i>	Arrays and vectors Declaration, Initialization, Accessing elements Example: Deck of cards Examples: Shuffle an array, Coupon collector, Selection sort	Homework 2 due
Week 6 <i>Sep 23 – Sep 29</i>	2D Arrays Example: Self-avoiding random walks	Homework 3 out
Week 7 <i>Sep 30 – Oct 6</i>	Input / Output Standard input/output streams Graphics libraries Examples: Function-plots, Fractals, Animations	Homework 3 due
<i>Oct 7 – Oct 13</i>	Midterm Examinations	
<i>Oct 14 – Oct 20</i>	Semester Break	Homework 4 out
Week 8 <i>Oct 21 – Oct 27</i>	Functions Arguments, Return values Pass by value/reference Example: Audio data Libraries, Modular programming	Homework 4 due
Week 9 <i>Oct 28 – Nov 3</i>	Memory Automatic (stack) and dynamic (heap) memory Pointers, References, Dynamic memory allocation Memory leak, dangling pointers	Homework 5 out

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Table 1: Weekly schedule of topics and assessments (Continued)

	Topic(s)	Notes
Week 10 <i>Nov 4 – Nov 10</i>	Recursion Base case, Reduction step Example: Tower of Hanoi	Homework 5 due
Week 11 <i>Nov 11 – Nov 17</i>	Structures The dot operator Example: Student database Linked lists	Homework 6 out
Week 12 <i>Nov 18 – Nov 24</i>	Classes Member functions, Constructors, Destructors Example: <code>std::string</code> ADT	Homework 6 due
Week 13 <i>Nov 25 – Dec 1</i>	Creating Data Types Example: Complex numbers, Fractional numbers	
Week 14 <i>Dec 2 – Dec 8</i>	Operator overloading	
Week 15 <i>Dec 9 – Dec 15</i>	Revision	Final lab exam

2.2 References

Following are useful references:

- *Introduction to C++* by George Tselikis, CRC Press, 2023.
- *C++ How to Program* by Paul Deitel and Harvey Deitel, 10/e, 2016.
- *Modern C++ for Absolute Beginners: A Friendly Introduction to the C++ Programming Language and C++11 to C++23 Standards* by Slobodan Dmitrovic, Apress, 2nd Edition, 2023.
- *C++ Primer* by Lippman, Lajoie, and Moo, 5th Edition, Addison-Wesley, 2012.
- *Programming: Principles and Practice Using C++* by Bjarne Stroustrup, 3rd Edition, Addison-Wesley, 2024.
- *Tutorials: C++ Language* at <https://cplusplus.com/doc/tutorial/>.

3 Logistics

3.1 Staff and Office Hours

 Farhan Siddiqui <i>Instructor</i> fsiddiqui@iba.edu.pk 📍 tba 📅 Office hours: tba	 Imran Rauf <i>Instructor</i> irauf@iba.edu.pk 📍 Room 206, Tabba Academic Block, Main Campus 📅 Office hours: Tue/Thu/Sat 11:30am–1pm
 Maria Rahim Khowaja <i>Instructor</i> mrkhowaja@iba.edu.pk 📍 Room 112 Faculty Lounge, Faisal Building, City Campus 📅 Office hours: Tue/Thu 03:00pm–05:00pm	 Samreen Kazi <i>Instructor</i> sakazi@iba.edu.pk 📍 Room 112 Faculty Lounge, Faisal Building, City Campus 📅 Office hours: Tue/Thu 03:00pm–05:00pm
 Mehwish Zafar <i>Instructor</i> mzafar@iba.edu.pk 📍 Room 22, NBP Building, Main Campus 📅 Office hours: tba	 Syed Taha <i>Teachig Assistant(Section: 97124)</i> s.taha.29208@khi.iba.edu.pk 📍 Main Campus 📅 Office hours: Wed:10:00am-2:15pm
 Piranchal <i>Teachig Assistant(Section: 97125)</i> p.ghai.29050@khi.iba.edu.pk 📍 Main Campus 📅 Office hours: Mon/Wed 1:00pm - 2:30pm	

3.2 Schedule

Table 2: Classes/labs schedule

📖 **Theory 97143**
 *Farhan Siddiqui*
📍 *C-8, Aman Tower*
📅 Tue/Thu: 2:30–3:45pm

📖 **Lab 97143**
 *Farhan Siddiqui*
📍 *CFL-6, Faisal Academic Block*
📅 Fri: 4:00–6:45pm

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Table 2: Classes/labs schedule (Continued)

 Theory 97116  <i>Imran Rauf</i>  <i>C-1, Aman Tower</i>  Tue/Thu: 8:30–9:45pm	 Lab 97116  <i>Imran Rauf</i>  <i>CCL-1, HBL Academic Block</i>  Sat: 8:30–11:15am
 Theory 97121  <i>Imran Rauf</i>  <i>C-1, Aman Tower</i>  Tue/Thu: 10:00–11:15am	 Lab 97121  <i>Mehwish Zafar</i>  <i>CCL-4, HBL Academic Block</i>  Sat: 11:30–2:15pm
 Theory 97141  <i>Maria Rahim</i>  <i>CFC-3, Faisal Academic Block</i>  Tue/Thu: 11:30–12:45pm	 Lab 97141  <i>Maria Rahim</i>  <i>CCL-1, HBL Academic Block</i>  Sat: 2:30–5:15pm
 Theory 97126  <i>Maria Rahim</i>  <i>CFC-3, Faisal Academic Block</i>  Tue/Thu: 1:00–2:15pm	 Lab 97126  <i>Maria Rahim</i>  <i>CCL-1, HBL Academic Block</i>  Sat: 11:30–2:15pm
 Theory 97125  <i>Samreen Kazi</i>  <i>CFC-4, Faisal Academic Block</i>  Tue/Thu: 11:30–12:45pm	 Lab 97125  <i>Samreen Kazi</i>  <i>CFL-6, Faisal Academic Block</i>  Sat: 2:30–5:15pm
 Theory 97124  <i>Samreen Kazi</i>  <i>CFC-4, Faisal Academic Block</i>  Tue/Thu: 1:00–2:15pm	 Lab 97124  <i>Samreen Kazi</i>  <i>CFL-6, Faisal Academic Block</i>  Sat: 10:00–12:45pm

4 Administration

4.1 Assessments

We have several assessment instruments designed for this course namely, homework, quizzes, midterm and final exams. All of these tools are important for learning the concepts we discuss in the class. The ‘personal struggle’ you engage in with these homework problems, quizzes, project, and exams will allow you to develop the skills necessary for success as a computer scientist. As a student in this class it is expected that you will always spend some time thinking about the problems/questions (on homework and labs) on your own before asking for hints, looking up solutions etc. Do not go in search of complete solutions online; learning the material happens when you are working on problems rather than looking up complete solutions.

You are welcome to collaborate on homework, provided that (1) you write up your solutions *individually*, and (2) you clearly *cite the names* of all collaborators and sources. Failure to do so will result in **zero credit**. An additional key requirement is that you should be able to explain what you submit. Inability to do so will result again in zero credit.

4.2 Grading

A *custom/relative* grading scale will be used based on overall performance of the class and difficulty-level of the class assessments as judged by the instructors. The final grade will be based on the following components:

Quizzes	15%
Homeworks	10%
Lab work	10%
Lab exam	10%
Midterm exam	20%
Final exam	35%

4.3 Late Work and Makeup Policy

No late solutions will be accepted and no make-up for exams or any of the quizzes will be given. If you have a valid medical excuse (for any of the quizzes, homework, midterm exam, viva etc.), the percentage of your grade corresponding to the missed work will be shifted to the next exam. Valid excuses require supporting documentation from a doctor (standard IBA policy will prevail in all cases of conflicts).

4.4 Attendance Policy

IBA attendance policy applies.

4.5 Academic Integrity

Each student in this course is expected to abide by the IBA Code of Conduct. Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by IBA regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- **PLAGIARISM:** Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. Plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
- **CHEATING:** The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
- **COLLUSION:** Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so. Any student violating academic integrity a second time in this course will receive a failing grade for the course, and additional disciplinary sanctions may be administered.

4.6 FAQs

- **Q.** What type of question will be asked in exam or quiz?
A. See previous exams / quizzes on LMS.
- **Q.** Would you do relative grading?
A. A custom grading scale will be used based on instructor's assessment of exam difficulty level and overall class performance.
- **Q.** I have missed the quiz because of xyz reason.
A. Best $n - 1$ quizzes would be considered in the final grade.
- **Q.** Syllabus for midterm, final and lab exams?
A. Everything discussed in class / lab / practice problems / homeworks.
- **Q.** Please extend the deadline for assignment!
A. No.

5 Course Outcomes

On completion of the course the student should have the following learning outcomes defined in terms of knowledge and skills:

- **Knowledge**

The student

- knows the principles of algorithmic thinking and programming
- understands how computer programming may be used to solve problems

- **Skills**

The student is able to

- construct small programs, using the programming languages covered, that include the use of arrays, functions and procedures, parameter passing with call by value and call by reference, custom data types, pointers, and references
- use modular and functional decomposition to break problems down functionally, and implement these structures in code as functions and procedures
- make use of available program libraries
- express functionality in terms of abstract data type or application programming interface
- use text-based tools of program development, including an editor, command-line tools, and a version control system

The following table lists the course learning outcomes (CLO's) for this course.

<i>Course Learning Outcomes (CLO's)</i>	
	Algorithmic Design and Implementation:
CLO-1	Students will be able to design, implement, and analyze basic algorithms, applying fundamental programming constructs such as loops, conditionals, and recursion to solve computational problems.
	Modular Program Development:
CLO-2	Students will demonstrate the ability to break down complex problems into smaller, manageable functions and modules, ensuring code reusability and clarity through the use of custom functions, procedures, and modular programming techniques.
	Effective Use of Development Tools:
CLO-3	Students will gain proficiency in using integrated program development tools, including editors, compilers, and version control systems, to write, debug, test, and document programs efficiently.

The mapping of course learning outcomes (CLO's) to program learning outcomes (PLO's) is as follows:

	Academic Education	Knowledge for Solving computing Problems	Problem Analysis	Design / Development of Solutions	Modern Tool Usage	Individual and Team Work	Communication	Computing Professionalism and Society	Ethics	Life-long Learning
	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO-10
CLO-1		✓								
CLO-2			✓							
CLO-3				✓						

The following table identifies the assessment instruments which will be used to assess the CLO's.

	CLO-1	CLO-2	CLO-3
	Algorithmic Design and Implementation	Modular Program Development	Effective Use of Development Tools
Labs		✓	
Homeworks		✓	✓
Quizzes	✓		
Exams	✓		✓

For reference, the graduate attributes (program learning outcomes - PLO's) taken from <https://www.seoulaccord.org/document.php?id=79> are as follows.

- **PLO-1. Academic Education**

[Educational depth and breadth]

Completion of an accredited program of study designed to prepare graduates as computing professionals

- **PLO-2. Knowledge for Solving Computing Problems**

[Breadth and depth of education and type of knowledge, both theoretical and practical]

Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements

- **PLO-3. Problem Analysis**

[Complexity of analysis]

Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines

- **PLO-4. Design / Development of Solutions**

[Breadth and uniqueness of computing problems, i.e., the extent to which problems are original and to which solutions have previously been identified or codified]

Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

- **PLO-5. Modern Tool Usage**

[Level and appropriateness of the tool to the type of activities performed]

Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations

- **PLO-6. Individual and Team Work**

[Role in, and diversity of, the team]

Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings

- **PLO-7. Communication**

[Level of communication according to type of activities performed]

Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions

- **PLO-8. Computing Professionalism and Society**

[No differentiation in this characteristic except level of practice]

Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice

- **PLO-9. Ethics**

[No differentiation in this characteristic except level of practice]

Understand and commit to professional ethics, responsibilities, and norms of professional computing practice

- **PLO-10. Life-long Learning**

[No differentiation in this characteristic except level of practice]

Recognize the need, and have the ability, to engage in independent learning for continual

development as a computing professional